

# **PROJECT PROPOSAL: MECHATRONICS SYSTEM INTEGRATION**

## **TITLE**

SMART WASHING MACHINE

## **AUTHORS**

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## **ABSTRACT**

This paper presents the design and development of a Smart Washing Machine system intended to optimize laundry safety and user interaction through a unified sensor integration and mobile connectivity framework. The proposed system is built around a single ESP32 microcontroller that serves as the central processing hub for all operational and safety functions. A key feature is the integration of an MPU6050 motion sensor to monitor machine stability, enabling automatic shutdown or user notification in the event of excessive vibration. User interaction is facilitated through a local control panel consisting of an LCD interface, a potentiometer for mode selection, and dedicated start/stop controls. Furthermore, the system includes a water level sensor to prevent overflows and provide overload notifications via a connected mobile device. The integration of these components into a single, cohesive system demonstrates a responsive mechatronic solution that enhances the reliability and convenience of modern domestic appliances.

## **1. INTRODUCTION**

Standard household laundry processes often face challenges regarding mechanical stability and user feedback. Conventional washing machines typically lack advanced diagnostic features to detect when a load has become dangerously unbalanced, which can lead to physical damage or excessive noise. Furthermore, users are often limited to fixed presets without intuitive real-time feedback on the machine's status. There is an increasing need for appliances that not only perform their primary function but also provide safety monitoring and remote accessibility.

Studies have shown that integrating inertial measurement units (IMUs) like the MPU6050 can significantly improve the safety of rotating machinery by detecting abnormal motion patterns. Additionally, providing users with a clear visual interface (LCD) and tactile controls (potentiometer and push buttons) improves the overall ergonomics of the appliance. Real-world incidents, such as machines "walking" across a floor due to imbalance or water overflowing due to sensor failure, highlight the necessity for a more integrated sensing approach.

To address these problems, this project proposes a unified smart control system that monitors stability, water levels, and wash modes simultaneously. The system uses an ESP32 to process inputs from a potentiometer for mode selection and a water level sensor for volume control. Feedback is provided through a dual-buzzer system and status LEDs, ensuring the user is alerted to the machine's state both locally and via a smartphone application.

**Table 1: Table of comparison**

| Ref/year | Project name                | Advantages                                   | Disadvantages                                | Comparison to my proposal                                                                                    |
|----------|-----------------------------|----------------------------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| [1]/2020 | Automatic Washer Controller | Robust motor control using PLC               | No remote monitoring or stability alerts     | My project replaces PLC with a single ESP32 for wireless connectivity and stability monitoring.              |
| [2]/2021 | IoT Washing Monitor         | Basic Wi-Fi notifications for cycle end      | Lacks physical UI for mode selection         | My project includes a potentiometer and LCD for local mode selection alongside IoT features.                 |
| [3]/2022 | Vibration Detection System  | High accuracy in detecting mechanical faults | No water level management or user control    | My project combines motion sensing with water level sensors and a full user control interface in one system. |
| [4]/2023 | Water-Saving Smart Machine  | Precision water level control                | Lacks local status indicators (LEDs/Buzzers) | My project adds buzzers and LEDs to provide immediate audio-visual feedback to the user.                     |

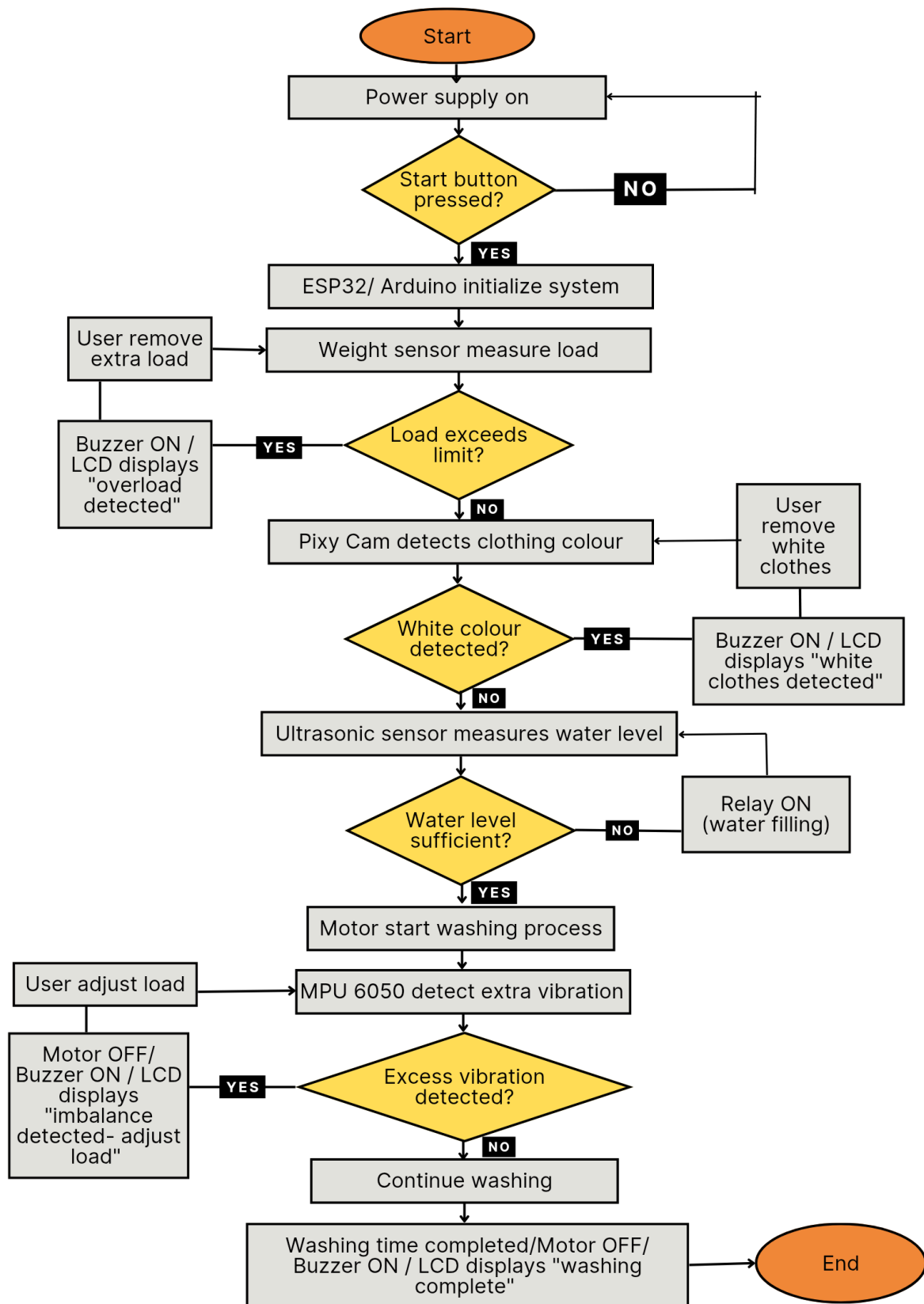
## **2. METHODOLOGY**

### **2.1 Project Development**

The Smart Washing Machine system is designed as a centralized mechatronic unit, where a single ESP32 microcontroller manages all input and output operations. The project development focuses on the integration of three core functional areas: Safety Monitoring, User Interface, and Automated Actuation.

The system integrates the MPU6050 motion sensor, which continuously tracks the machine's stability during the spin cycle. If the machine becomes unstable or tilted, the ESP32 is programmed to automatically cut power to the DC motor or notify the user via the mobile app. Simultaneously, the water level sensor ensures the machine does not exceed capacity, preventing overflows and detecting potential overloads.

For user interaction, the system utilizes a potentiometer to allow the user to navigate through different wash modes, which are displayed in real-time on an LCD. Two push buttons provide dedicated "Start" and "Stop" functions. Status indication is handled by two LEDs (Ready and In-Action), while two buzzers provide audible signals at the start and completion of the spinning process. This unified architecture ensures that all sensor data is processed instantly by a single controller, reducing latency and complexity in the system design.

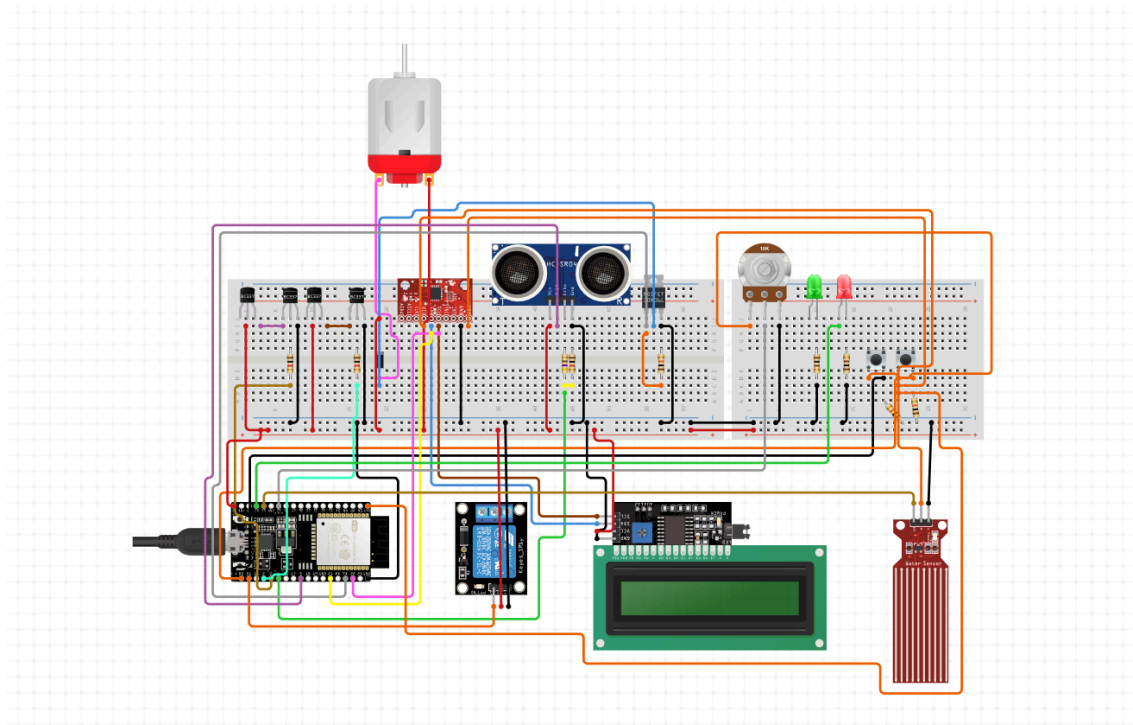


**Figure 1.** Flowchart

## 2.2 Schematic Diagram and Prototype Setup

The electrical design centers on the ESP32 as the primary controller for the entire machine. The hardware setup includes the MPU6050 and water level sensor positioned to monitor physical conditions directly. The DC motor, which drives the drum, is controlled via a driver circuit to allow for varying speeds according to the selected mode.

The control panel components—the potentiometer, LCD, and push buttons—are connected directly to the ESP32's GPIO pins. The status LEDs and buzzers are strategically placed to provide immediate sensory feedback. By utilizing the ESP32's Wi-Fi capability, the system bridges these local hardware components with a mobile application, allowing for remote monitoring and receiving "Overload" or "Unstable" notifications. This single-system design provides a comprehensive mechatronic solution that modernizes the laundry experience through efficient integration.



**Figure 2.** Wiring diagram